

ABU DHABI BATEEN, United Arab Emirates

WMO: 412160

Lat: 24.428N

Lon: 54.458E

Elev: 16

StdP: 14.69

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.2	59.1	33.7	28.4	88.2	38.3	34.0	86.7	21.1	69.3	19.0	69.9	3.2	150	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.1	112.5	74.1	109.6	74.6	107.5	75.0	87.8	95.0	86.9	94.5	86.0	94.1	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
86.2	191.5	92.8	84.6	182.1	92.5	84.1	178.9	92.5	53.0	95.0	51.9	93.9	50.8	93.9	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	17.6	15.7	53.6	116.4	1.7	1.7	52.4	117.6	51.4	118.5	50.4	119.5	49.1	120.7
			47.4	89.7	2.7	0.7	45.5	90.2	43.9	90.6	42.4	91.0	40.4	91.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	84.3	68.3	70.2	75.1	83.2	91.2	94.5	98.2	98.8	94.0	87.3	78.8	71.3	(6)
(7)		DBStd	11.51	3.10	4.91	5.51	5.72	3.97	3.33	3.93	2.95	3.04	3.36	3.65	3.36	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	10	5	4	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD50	12529	566	567	779	996	1276	1336	1493	1513	1321	1158	865	659	(10)
(11)		CDD65	7065	107	151	314	546	811	886	1028	1048	871	693	415	195	(11)
(12)		CDH74	96560	364	827	2231	6211	11873	13854	16758	16916	13390	9430	3875	831	(12)
(13)		CDH80	61920	56	273	853	3134	7573	9543	12296	12452	9081	5245	1313	101	(13)
(14)	Wind	WSAvg	7.3	6.6	8.0	8.4	7.6	7.6	7.7	7.6	7.6	7.1	6.4	6.5	6.4	(14)
(15)	Precipitation	PrecAvg	2.4	0.4	0.4	0.6	0.2	0.1	0.0	0.1	0.1	0.0	0.0	0.1	0.4	(15)
(16)		PrecMax	6.3	2.0	3.6	2.3	2.1	1.2	0.8	0.7	1.1	0.1	0.3	1.0	2.5	(16)
(17)		PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.8	0.7	0.7	0.7	0.5	0.2	0.2	0.2	0.2	0.0	0.0	0.2	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	93.1	98.7	105.8	111.3	112.7	115.2	114.6	110.3	102.3	95.0	86.0	(19)
(20)		MCWB	63.0	63.6	64.2	67.7	70.5	71.8	74.5	74.1	72.6	71.7	66.8	66.3	(20)	
(21)		2%	DB	81.0	87.4	93.1	101.9	107.6	109.1	112.8	111.5	106.5	99.0	91.4	82.5	(21)
(22)		MCWB	62.5	63.2	64.9	67.9	70.2	73.3	74.9	74.9	74.0	73.2	69.6	67.5	(22)	
(23)		5%	DB	78.1	82.8	88.2	97.8	104.3	105.9	109.5	109.3	104.0	97.0	88.7	80.6	(23)
(24)		MCWB	62.9	62.8	64.9	68.1	71.0	74.4	75.8	75.8	75.1	73.8	70.3	67.7	(24)	
(25)		10%	DB	75.1	78.8	84.3	93.5	100.8	103.6	106.1	106.2	101.2	94.9	86.3	78.6	(25)
(26)		MCWB	63.3	63.3	65.2	68.1	71.8	75.3	77.5	77.1	77.1	74.7	70.5	66.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	70.2	72.8	77.9	82.8	87.1	89.0	89.0	87.5	84.3	77.4	72.9	(27)
(28)		MADB	74.7	77.4	81.1	85.9	91.7	94.5	96.0	95.4	93.0	90.5	84.9	78.1	(28)	
(29)		2%	WB	67.4	68.8	71.5	76.0	81.0	85.7	87.7	88.0	86.3	82.7	75.6	71.7	(29)
(30)		MADB	73.4	75.4	79.3	85.4	89.6	93.5	95.5	95.0	92.9	89.5	83.8	78.1	(30)	
(31)		5%	WB	66.2	67.4	70.1	74.3	79.6	84.5	86.5	87.2	85.4	81.5	74.5	70.2	(31)
(32)		MADB	72.8	74.4	77.7	85.1	89.0	93.0	94.9	94.7	92.6	89.0	83.0	76.5	(32)	
(33)		10%	WB	65.0	66.1	69.0	72.7	78.2	83.3	85.6	86.3	84.4	80.3	73.3	69.1	(33)
(34)		MADB	72.1	74.2	77.0	84.3	88.8	92.3	94.6	94.5	92.2	88.6	82.2	75.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.8	15.4	16.4	19.5	21.1	19.9	18.5	19.1	17.9	17.2	15.3	14.3	(35)	
(36)		5% DB	MADB	19.8	21.9	22.5	25.0	25.7	23.5	22.3	21.9	20.7	19.0	17.7	16.5	(36)
(37)		MCWBR	9.0	9.6	9.7	11.2	11.4	12.2	13.3	13.8	12.8	10.4	8.6	7.9	(37)	
(38)		5% WB	MADB	16.4	17.0	16.6	22.1	21.8	20.1	18.4	19.0	18.1	17.1	16.4	15.0	(38)
(39)		MCWBR	8.7	8.9	8.6	10.4	10.5	10.6	11.3	11.6	10.6	9.2	8.0	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.454	0.506	0.572	0.611	0.596	0.634	0.727	0.700	0.598	0.508	0.463	0.444	(40)	
(41)		taud	2.035	1.891	1.740	1.689	1.706	1.669	1.555	1.606	1.755	1.930	2.050	2.068	(41)	
(42)		Ebn at Noon	250	245	236	230	233	223	203	208	226	241	245	247	(42)	
(43)		Edh at Noon	49	60	73	78	77	79	89	84	71	57	47	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1331	1571	1846	2107	2334	2376	2229	2155	2007	1755	1426	1249	(44)	
(45)		RadStd	73	124	123	125	65	48	71	44	40	30	36	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.90	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	+297	+288

Nomenclature: See separate page